SUM OF NATURAL NUMBERS

1. Sum of first ` n ‘ Natural numbers: $\sum N = \frac{n(n+1)}{2}$
2. Sum of ` n ‘ Natural numbers $= \frac{n(a+l)}{2}$
3. Sum of squares of first ` n ‘ Natural numbers $\sum N^2 = \frac{n(n+1)(2n+1)}{6}$
4. Sum of cubes of first ` n ‘ Natural numbers $\sum N^3 = \frac{n^2(n+1)^2}{4} = \left(\frac{n(n+1)}{2}\right)^2 = (\sum N)^2$

EXAMPLES:-

1. **1+2+3+4+----------+10 = -------**

2. **1+2+3+4+----------+25 = -------**

3. **1+2+3+4+----------+50 = -------**

4. **1+2+3+4+----------+100 = -------**

5. **1+2+3+4+----------+110 = -------**

6. **1+2+3+4+----------+250 = -------**

7. **1+2+3+4+----------+500 = -------**

8. **1+2+3+4+----------+1000 = -------**

9. **1+2+3+4+----------+700 = -------**

10. **1+2+3+4+----------+950 = -------**

11. **11+12+13+14+----------+50 = -------**

12. **11+12+13+14+----------+100 = -------**

13. **10+11+12+13+14+----------+50 = -------**

14. **72+73+74+----------+88 = -------**

15. **11+12+13+14+----------+50 = -------**

16. **112+113+114+----------+550 = -------**

17. $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + - - - - - - - - + 10^2 = - - - - -$

18. $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + - - - - - - - - + 20^2 = - - - - -$

19. $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + - - - - - - - - + 18^2 = - - - - -$

20. $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + - - - - - - - - + 25^2 = - - - - -$

21. $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + - - - - - - - - + 50^2 = - - - - -$

22. $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + - - - - - - - - + 100^2 = - - - - -$

23. $1^3 + 2^3 + 3^3 + 4^3 + 5^3 \pm - - - - - \mp 10^3 = - - - - -$

24. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 24^3 = - - - - -}$

25. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 25^3 = - - - - -}$

26. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 60^3 = - - - - -}$

27. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 72^3 = - - - - -}$

28. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 90^3 = - - - - -}$

29. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 100^3 = - - - - -}$

30. $\mathbf{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + - - - - - - - - + 120^3 = - - - - -}$

31.